INCH-POUND

MIL-S-70437 (AR)
AMENDMENT 3
29 September 1994
SUPERSEDING
AMENDMENT 2
2 December 1985

MILITARY SPECIFICATION

SPITBACK LEAD ASSEMBLY
FOR
FUZE, POINT DETONATING, M759

This amendment forms a part of MIL-S-70437 (AR), dated 15 July 1983, and is approved for use by the US Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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* Add the following paragraph:

"3.3.3 Ballistic performance. The spitback lead assembly, assembled into a M759 fuze/M789 cartridge (fuze carrier), shall upon impact with a 0.063 inch (nominal thickness) 2024-T3 aluminum alloy plate at a range of 150 meters minimum, cause the projectile to produce an average hole of not less than seven (7) inches in diameter, exclusive of petals, and show evidence of shrapnel on the forward face or remnants of the aluminum plate."

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4.3.1: Delete "Leas Assemblies, 11834355" and substitute "Lead Assemblies 11834355".

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Table 1:

Delete “Load Assembly” and substitute "Lead Assembly".

Functioning - sensitivity and output, delete "Note 1" and substitute "Note 2".

Change 3.5.1 to 3.3.1

Functioning - varicomp initiation

Change 3.5.2 to 3.3.2

NOTE 3 delete "Table" add Table

for n = 4, a = 0, r = 9** (minimum sample)
n = 6, a = 1, r = 9
n = 8, a = 2, r = 9
n = 10, a = 3, r = 9
n = 12, a = 4, 4 = 9
n = 14, a = 5, r = 9
n = 16, a = 6, r = 9
n = 18, a = 7, r = 9
n = 20, a = 8, r = 9 (maximum sample)

** In this table "r=q” refers to the maximum number of rejects allowed before rejecting the lot. Less than 9 rejects allows testing to continue until the lot passes or the maximum sample (n) condition is reached.

Add the following to Table 1:

“Ballistic Performance 78 ACC-1/REJ-2 3.3.3 4.5.2.3”

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4.4.2: Delete "wtith" and substitute "with".

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4.4.2.1, Major 101: Delete “3.2” and substitute "4.4”.

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4.4.2.2, Major 101 thru 107: Delete “3.2" and substitute “4.4”.

Minor 201: Delete "3.2 and 3.6” and substitute "3.4”.

Add: "Minor 202 workmanship, .65, 3.4.”

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Notes: Delete "n = 21, a = 8, r = 9 (minimum sample)” and substitute “n = 20, a = 8, r = 9 (maximum sample)".

* 4.4.2.3: Delete “101. Sensitivity and output 0.40% 3.5.1 and 4.5.2.1” and substitute “101. Sensitivity and output 0.40% 3.5.1 4.5.2.1/4.5.2.3”.

Major 102: Delete "3.5.2” and substitute “3.3.2”.

Note 1 Delete “Table” add “Table”

for n = 4, a = 0, r = 9** (minimum sample)
n = 6, a = 1, r = 9
n = 8, a = 2, r = 9
n = 10, a = 3, r = 9
n = 12, a = 4, r = 9
n = 14, a = 5, r = 9
n = 16, a = 6, r = 9
n = 18, a = 7 r = 9
n = 20, a = 8 r = 9

** In this table "r=q" refers to the maximum number of rejects allowed before rejecting the lot. Less than 9 rejects allows testing to continue until the lot passes or the maximum sample (n) condition iS reached.

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* 4.5.2.1: Delete in its entirety and substitute and following:

“4.5.2.1 Sensitivity and output. The sample spitback lead assemblies shall be randomly selected from each lot. The lead shall be assembled in accordance with drawing 11834346, except that the explosive pellet and explosive holder shall be omitted and the spitback standoff test holder (11834349) placed directly against the witness plate (11834351). A weight of

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0.25 ounce plus or minus 0.01 ounce shall be dropped on the firing pin, to function the detonator, from a height of five (5) inches minimum. A lead that falls to fire or fires but fails to produce a dent in the steel witness block of the required depth shall be classed defective if the detonator confinement sleeve expansion is 0.330 inch diameter or greater. If the confinement sleeve expansion is less than 0.330 inch diameter, the lead shall not be classed defective, but the detonator calssed defective and the result a no-test. The indentation depth shall be measured using a probe with a 0.017 inch plus or minus 0.002 inch spherical radius tip.

* Add new paragraph 4.5.2.3 as follows:

"4.5.2.3 Ballistic performance. The spitback lead assembly, assembled into a M759 fuze/M789 cartridge (fuze carrier), shall within two (2) minutes after removal from a temperature conditioning chamber (21°C) be tested at a range of 200 meters ± 20 meters with an angle of impact of zero (0) degrees ± 10 degrees on 0.063 inch 2024-T3 aluminum plate. Failure of the spitback to effect projectile high order function and produce a hole as defined in paragraph 3.3.3 shall be considered a defect. Any defect that is determined to be caused by a defective fuze, defective cartridge, or defective weapon, shall be considered a no-test."

* Add new paragraph 4.5.2.4 as follows:

"4.5.2.4 Combined testing of detonator and spitback lead (Optional). If detonators and spitback leads are being produced by the same manufacturer, the sensitivity and output test of this specification and the sensitivity and output test of MIL-D-70436 may be combined. A detonator that fails to fire shall be classed "defective". A detonator that fires but fails to initiate the lead or to produce a dent of the required depth in the steel witness block shall be classed "defective" if the detonator confinement sleeve expansion is less than 0.330 inch diameter; should this occur, the lead under test should be declared a "no-test". A lead that fails to fire, or fires but fails to produce a dent of the required depth in the steel witness block shall be classed "defective" if the detonator confinement sleeve expansion is 0.330 inch diameter or greater; should this occur, the detonator under test shall be declared "acceptable". Spare units will be tested as needed to ensure that the required sample size is tested for each component in the combined test.

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If the total number of defective detonators exceeds the allowable number of defects (per MIL-D-70436) then the detonator lot shall be rejected. If the total number of defective leads exceeds the allowable number of defects, then the lead lot shall be rejected. If the total number of defective detonators exceeds the allowable number of defects (per MIL-D-70436) and the total number of defective leads is at the maximum allowable for the sample, the detonator lot shall be rejected and the lead lot shall be declared a “no-test. Retest of a rejected detonator lot, after corrective action, shall be per MIL-D-70436. Retest of a rejected lead lot, after corrective action, shall be per 4.5.2.1 of this specification. If both the detonator and lead lots are rejected, retest of both lots may be per this paragraph, after corrective action."

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